

KOLEJ MATRIKULASI KELANTAN
DC014 COMPUTER SCIENCE
TRIAL PSPM 2024/2025
(2 HOURS/70 MARKS)

Name : _____ Class: _____

Matric No : _____

1. In today's business environment, multimedia plays a crucial role in communication, training, and presentations. Professionals across all sectors use multimedia elements to enhance and improve their work. Ahmad, a corporate trainer, uses graphics in his business presentations to deliver his content effectively.

- a. Describe one (1) other multimedia element that can be used in a business training presentation besides graphics.

[2 marks]

.....
.....

- b. Explain one (1) importance of using the multimedia element you mentioned in 1(a) in a business training context.

[2 marks]

.....
.....

- c. In the business world, multimedia is frequently used in presentations, product demonstrations, and training materials. Different types of multimedia representations can convey information in various ways.

Differentiate between the axes used in two-dimensional (2D) and three-dimensional (3D) multimedia representations in the context of business presentations.

[2 marks]

.....
.....

- d. Multimedia delivery plays a crucial role in distributing business training content. Two (2) common types of multimedia delivery are online and offline.

Choose a type of multimedia delivery that is suitable for delivering business training content. Based on the option you choose, justify the advantages of your choice.

[2 marks]

.....

.....

- e. Explain an application of multimedia in the following field:

i. Entertainment

[2 marks]

.....

.....

ii. Healthcare

[2 marks]

.....

.....

- f. Navigation provides a structure for linking different sections of your content, helping you organize and present your information effectively.

i. Name the two (2) types of navigation commonly found on websites.

[2 marks]

.....

.....

ii. Describe a type of navigation mentioned in part 1(f)(i).

[2 marks]

.....

.....

2. Multimedia design involves combining various elements such as text, images, audio, and video to engage an audience.

a. The simplicity and effectiveness of the three-phase (3Ps) model have led to its wide use as a conceptual framework for the YouTube video production process.

i. Describe an activity in the production phase of creating a YouTube video.

[2 marks]

.....

.....

ii. Describe an activity in the post-production phase of a YouTube video creation.

[2 marks]

.....

.....

b. An authoring tool is a program that allows you to create digital content and export it to various formats.

Describe a type of multimedia authoring tool a YouTuber might use. [2 marks]

.....

.....

3. Computer technology has been used to facilitate the storage, production, and updating of data in various fields, including e-commerce.

a. Explain the concept of the database system in e-commerce.

[2 marks]

.....

.....

b. Describe the advantages of using a database system compared to a file processing system in e-commerce.

[2 marks]

.....

.....

- c. Identify the component of a database system in an e-commerce context based on the description below.

[2 marks]

Description	Component of Database System
i. The instructions and rules that govern the design and use of the database system, ensuring security, privacy, and integrity in handling customer data.	
ii. The raw data or 'material' from which useful information is generated, such as customer demographics, product details, and sales records.	

4. Data integrity is concerned with the accuracy, consistency and reliability of data stored in databases or other data storage systems.

- a. Describe the difference between data accuracy and data consistency in the context of data integrity.

[2 marks]

.....

.....

- b. Identify the database object based on the description below.

[2 marks]

Description	Database Object
i. An interactive interface that allows users to enter, edit, and view data in a database in a structured and user-friendly way.	
ii. A formatted output that organizes and presents data from the database, often used for analysis or sharing insights with others.	

5. a. Describe the several uses of the network for the following.

i. Monitoring Systems

[2 marks]

.....

.....

ii. Telemedicine

[2 marks]

.....

.....

b. In a network, computers and devices connected together via transmission media and communication devices.

i. Explain wired transmission media?

[2 marks]

.....

.....

ii. Identify type of transmission media where no physical medium is required for the transmission of electromagnetic signals.

[1 mark]

.....

.....

iii. State two examples of transmission media in question 5(b)(ii) that is appropriate for office network

[2 marks]

.....

.....

c. i. Describe network communication devices?

[2 marks]

.....

.....

- ii. Name and describe the most appropriate communication device X in
FIGURE 1.

[2 marks]

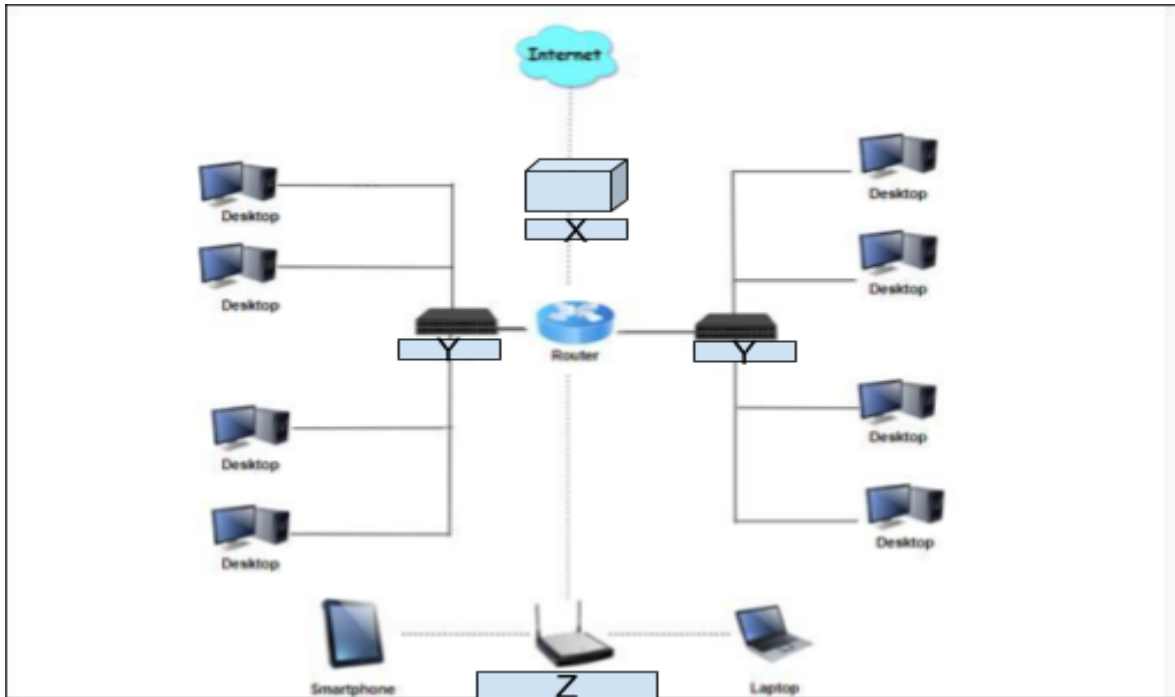


FIGURE 1

-
-
- iii. Identify the most appropriate communication device to replace
communication device Y if the user wants to opt a cheaper option

[1 mark]

-
- iv. State two (2) communication devices other than communication device X,
communication device Y, router and communication device Z in FIGURE 1.

[2 marks]

.....

.....

- v. Describe the function of communication device base on one of your answer in 5(c)(iv) [2 marks]

.....

.....

6. a. Internet searching refers to the process of using the Internet to locate and retrieve information from the vast resources available on the World Wide Web. State one characteristic for each type of search tools below.

[2 marks]

Type of search tool	Characteristic
i. Search engine	
ii. Subject directory	

- b. Identify the correct search strategies based on given scenario

[4 marks]

Scenario	Search Strategies
i. Nancy wants to find information about cats including kittens by using Google search.	
ii. Ahmad prefers to use this style of search strategy by typing more than one keyword in a keyword search box.	
iii. Kumar type site https://publicinfobanjir.water.gov.my/ to search the specific website by using Google search.	
iv. To further specify exactly what you want a search engine to find by using AND, OR and NOT.	

- c. Web activities can display content, catalog entries, or categories. When you create a web activity, the action element that you choose determines the type of information to display. Identify the web activities and the example based on the scenario given.

[6 marks]

Scenario	Web activities	Example
i. Nina likes to share their personal content, interests, activities, backgrounds or real-life connections.		

ii. As an influencer, Alex wants his followers to buy the products on online platform.		
iii. Azman used the web for listening the songs, watching video and movies.		

d. Censorship is the suppression of speech, public communication, or other information. This may be done on the basis that such material is considered objectionable, harmful, sensitive, or "inconvenient". Based on your understanding, describe one purpose of censorship. [2 marks]

.....

.....

e. Data privacy is the relationship between the collection and dissemination of data, technology, the public expectation of privacy, and the legal and political issues surrounding them. Suggest two ways on how to safeguard our personal information while web browsing.

[2 marks]

.....

.....

f. The Internet of Things (IoT), which has been called the core of 4IR, is a network of sensors in items that generates data on how they are used.

i. Describe one (1) risk of the application of 4IR technology.

[2 marks]

.....

.....

ii. Describe one (1) example of Internet of Things (IoT) that can be used at home.

[2 marks]

.....

.....

END OF QUESTION PAPER